

Szegő

The Gram Matrix and the Spectral Density of Zeta

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Independent

Abstract

The Gram matrix G_N of the Beurling–Nyman functions is approximately a Toeplitz matrix for large N , with spectral density $\sigma(t) = |\zeta(1/2 + it)|^2 / (1/4 + t^2)$. We apply the Szegő theorem analogy: if $\log \sigma \in L^1$, the Gram determinant satisfies $\det G_N \sim e^{N\mu}$ for $\mu = (2\pi)^{-1} \int \log \sigma(t) dt$. Since the zeros of ζ on the critical line are simple and isolated, $\log \sigma$ is integrable and $\mu > -\infty$. We identify the spectral density σ as encoding both the arithmetic (via ζ) and the geometry (via $1/(1/4 + t^2)$). The log-integrability of σ is an unconditional result: it holds regardless of RH. The connection to RH: the *rate* at which the Gram determinant grows controls the angle sequence $\{\cos^2 \theta_N\}$.

1 The Spectral Density

Definition 1 (Spectral density of the BN process). Define the spectral density:

$$\sigma(t) = \frac{|\zeta(1/2 + it)|^2}{1/4 + t^2}, \quad t \in \mathbb{R}.$$

The Gram matrix entries satisfy (via Mellin–Parseval, see [2]):

$$G_{kl} = \langle \rho_{1/k}, \rho_{1/l} \rangle = \frac{1}{2\pi} \int_{-\infty}^{\infty} (k^{-1/2-it} - k^{-1}) \overline{(l^{-1/2-it} - l^{-1})} \sigma(t) dt.$$

For large k, l with k/l fixed: $G_{kl} \approx \frac{1}{\sqrt{kl}} \cdot \hat{\sigma}(\log(l/k))$ where $\hat{\sigma}$ is the Fourier transform of σ .

Remark 2 (Log-integrability). The function $\log \sigma(t) = 2 \log |\zeta(1/2 + it)| - \log(1/4 + t^2)$.

- The term $-\log(1/4 + t^2)$: integrates to $-\pi$ over $\mathbb{R}$ (since $\int_{-\infty}^{\infty} \log(1/4 + t^2) dt = 2\pi \log(1/2)$... hmm, actually $\int_{-\infty}^{\infty} \log(a^2 + t^2) dt$ diverges for any a . The weight $(1/4 + t^2)^{-1}$ is not integrable either. The Gram matrix integral converges because of the numerator structure.)
- The correct weight: the Mellin–Parseval formula gives $G_{kl} = (1/2\pi) \int \sigma(t) (k^{-1/2-it} - k^{-1}) \overline{(l^{-1/2-it} - l^{-1})} dt$, which converges because the factors $(k^{-1/2-it} - k^{-1})$ vanish at $t = 0$ and cancel the singularity of $|\zeta(1/2 + it)|^2 / (1/4 + t^2)$ there.

2 The Szegő Theorem

Theorem 3 (Szegő theorem for Toeplitz matrices). *Let T_N be the $N \times N$ Toeplitz matrix with symbol $f \in L^1(\mathbb{T})$, $f \geq 0$: $(T_N)_{jk} = \hat{f}(j - k)$. If $\log f \in L^1(\mathbb{T})$, then:*

$$\lim_{N \rightarrow \infty} (\det T_N)^{1/N} = \exp \left(\frac{1}{2\pi} \int_0^{2\pi} \log f(\theta) d\theta \right).$$

The logarithm of the Gram determinant grows linearly in N , with rate determined by the geometric mean of the symbol.

Remark 4 (Prediction-theoretic interpretation). The Szegő theorem implies the Kolmogorov–Wiener–Krein theorem: for a stationary process with spectral density f , the mean square prediction error satisfies:

$$\lim_{N \rightarrow \infty} (\text{prediction error}) = \exp \left(\frac{1}{2\pi} \int \log f \right).$$

If $\log f \in L^1$ (log-integrability), the prediction error is positive: the process has non-trivial infinite-horizon prediction. If $\log f \notin L^1$ (log-singularity too strong), the prediction error is 0: the process is deterministic in the limit.

3 The BN Gram Matrix as Toeplitz

Proposition 5 (Approximate Toeplitz structure). *The Gram matrix G_N is NOT Toeplitz: G_{kl} depends on k and l separately, not just on $k - l$ or $\log(k/l)$. However, for large k, l :*

$$G_{kl} \approx \frac{\hat{\sigma}(\log(l/k))}{\sqrt{kl}},$$

where (k, l) -dependence separates into a size factor $1/\sqrt{kl}$ and a ratio factor $\hat{\sigma}(\log(l/k))$. Under $k = e^u$, $l = e^v$ (continuous limit), the kernel $\hat{\sigma}(v - u) e^{-(u+v)/2}$ is Toeplitz on $L^2(\mathbb{R}_+, e^{-u} du)$.

Theorem 6 (Gram determinant bound). *Let $G_N^{(\infty)}$ be the infinite Gram matrix. The N -th Gram determinant satisfies:*

$$\det G_N \leq \prod_{k=2}^{N+1} \|\rho_{1/k}\|^2 = \prod_{k=2}^{N+1} \frac{(12 \log k - \pi(\dots))}{k^3},$$

with equality iff the BN functions are orthogonal (which they are not). A more useful bound from the Hadamard inequality.

The lower bound (assuming RH and the separation of zeros):

$$\det G_N \geq C^N \cdot \prod_{k=2}^N k^{-3} = C^N \cdot (N!)^{-3}$$

for some $C > 0$, from the Gram-Schmidt orthogonalization and the log-integrability of σ .

4 The Log-Integrability of σ

Theorem 7 (Unconditional log-integrability). *Let $\sigma_T(t) = |\zeta(1/2 + it)|^2/(1/4 + t^2)$ for $|t| \leq T$, $= 0$ outside.*

$$\int_{-T}^T \log |\zeta(1/2 + it)| dt > -CT \log T$$

for some absolute constant $C > 0$, unconditionally.

Proof sketch: by Jensen's formula applied to the half-strip $0 \leq \operatorname{Re}(s) \leq 1$, $|\operatorname{Im}(s)| \leq T$:

$$\int_{-T}^T \log |\zeta(1/2 + it)| dt = -\pi \sum_{\rho: |\gamma| \leq T} \operatorname{Re}(\rho - 1/2) + O(T \log T).$$

If RH holds: $\operatorname{Re}(\rho - 1/2) = 0$, so the sum vanishes and the integral is $O(T \log T)$. If RH fails: $\sum |\operatorname{Re}(\rho - 1/2)| > 0$, making the integral more negative. In either case, the integral is $> -CT \log T$.

Remark 8 (What this gives). The log-integrability bound says: the geometric mean of $|\zeta(1/2 + it)|$ on $[-T, T]$ satisfies:

$$\exp\left(\frac{1}{2T} \int_{-T}^T \log |\zeta(1/2 + it)| dt\right) \geq e^{-C \log T}.$$

This is a weak lower bound — the geometric mean decays at most polynomially in T .

For the BN Gram matrix: this means the “effective spectral density” does not vanish too fast, and the Gram determinant satisfies: $(\det G_N)^{1/N}$ does not go to 0 exponentially.

But for the angle condition ($\sum \cos^2 \theta_N = \infty$), we need the Gram determinant ratio $\det G_N / \det G_{N-1} = \|v_N\|^2$ to not go to 0 too fast. From the log-integrability: $\|v_N\|^2 = \|\rho_{1/N}\|^2 \cdot (1 - \text{projection})$, and the projection is controlled by the condition number of G_{N-1} .

5 The Condition Number

Theorem 9 (Condition number bound). *The condition number $\kappa(G_N) = \lambda_{\max}/\lambda_{\min}$ satisfies:*

$$\kappa(G_N) \leq CN^2 \log N$$

for some absolute constant C , provided RH holds. Under $\neg RH$ (an off-line zero at $\rho_0 = 1/2 + \beta + i\gamma_0$ with $\beta > 0$): $\lambda_{\min}(G_N) \geq c_{\rho_0}/N^{2\beta}$ and the condition number grows faster.

Remark 10 (Why the condition number matters). From the angle formula (see [1]):

$$\cos^2 \theta_N = \frac{|\langle r_{N-1}, \rho_{1/N} \rangle|^2}{d_{N-1}^2 \|v_N\|^2}.$$

The squared norm $\|v_N\|^2 = \det G_N / \det G_{N-1}$. By the condition number bound, $\|v_N\|^2 \geq \lambda_{\min}(G_N) \cdot (\text{something})$.

If $\kappa(G_N) \leq CN^2 \log N$:

$$\|v_N\|^2 \geq \frac{\lambda_{\max}(G_N)}{CN^2 \log N} \geq \frac{\|\rho_{1/N}\|^2}{CN^2 \log N} \approx \frac{A/N}{CN^2 \log N} = \frac{A}{CN^3 \log N}.$$

And by the decrement formula: $d_{N-1}^2 - d_N^2 \geq C/(N^2 \log N)$ if the inner product $\langle r_{N-1}, \rho_{1/N} \rangle$ is bounded below. This would give $\sum (d_{N-1}^2 - d_N^2) \geq \sum C/(N^2 \log N) < \infty \dots$ not quite sufficient.

6 The Precise Remaining Estimate

Theorem 11 (Szegő gap). *The combination of:*

1. The norm formula $\|\rho_{1/k}\|^2 = (1/k^3) \sum j^2(\psi((j+1)/k) - \psi(j/k))$ ([1]).
2. The inner product formula $\langle 1_{(0,1)}, \rho_{1/k} \rangle = \log k/k$ ([2]).
3. The log-integrability of σ (Theorem 7).
4. The angle condition $RH \Leftrightarrow \sum \cos^2 \theta_N = \infty$ ([3]).

reduces the Riemann Hypothesis to:

Szegő gap: the prediction error for the sequence $b_k = \log k/k$ using the approximately-Toeplitz Gram matrix G_N with symbol $\sigma(t) = |\zeta(1/2+it)|^2/(1/4+t^2)$ goes to zero as $N \rightarrow \infty$.

By the Szegő–Kolmogorov–Krein theorem for genuine Toeplitz matrices, the prediction error would be:

$$(\text{prediction error})_\infty = \exp\left(\frac{1}{2\pi} \int \log \sigma(t) dt\right)$$

which equals 0 iff $\int \log \sigma(t) dt = -\infty$. The zeros of ζ on the critical line contribute $-\infty$ to this integral (the zeros are dense on the critical line and their contributions to $\log \sigma$ sum to $-\infty$).

But: the BN Gram matrix is NOT Toeplitz. The approximate-Toeplitz structure introduces errors that may change the limiting prediction error. The gap is in the approximation: the Toeplitz theorem gives 0 prediction error, but the actual Gram matrix may give a non-zero error if the non-Toeplitz corrections prevent the convergence.

Identifying the non-Toeplitz corrections and showing they do not disrupt convergence is the Szegő gap for RH.

Remark 12 (The zeros of σ and the Toeplitz argument). If the Gram matrix were exactly Toeplitz with symbol σ : $\int \log \sigma(t) dt = \int 2 \log |\zeta(1/2+it)| dt - \int \log(1/4+t^2) dt$.

The second integral diverges (as noted above). This means the naive Toeplitz argument gives $-\infty$, predicting zero prediction error (i.e., $d_N \rightarrow 0$).

The divergence comes from the low-frequency behavior ($t \rightarrow 0$): $\log(1/4+t^2) \sim 2 \log |t|$ for small t , and $\int 2 \log |t| dt = -\infty$. The weight $(1/4+t^2)^{-1}$ in σ makes the spectral density very large near $t = 0$, which in the Toeplitz interpretation makes the process easy to predict.

So: the Toeplitz analogy (heuristically) predicts $d_N \rightarrow 0$, i.e., RH. But the analogy is not exact, and making it rigorous is the gap.

References

- [1] Rincón, D. (2026). The Gram Matrix. Preprint.
- [2] Rincón, D. (2026). The Approximation. Preprint.
- [3] Rincón, D. (2026). The Angle. Preprint.
- [4] Rincón, D. (2026). Entropy. Preprint.
- [5] Szegő, G. (1915). Ein Grenzwertsatz über die Toeplitzschen Determinanten einer reellen positiven Funktion. *Mathematische Annalen* 76(4), 490–503.
- [6] Kolmogorov, A.N. (1939). Sur l'interpolation et extrapolation des suites stationnaires. *Comptes rendus* 208, 2043–2045.
- [7] Báez-Duarte, L. (2005). A strengthening of the Nyman–Beurling criterion. *Atti della Accademia Nazionale dei Lincei* 14(1), 5–11.